Prepared By	08-Aug-2012	Mr.Chatchawan S.	RNC RENEWCOILS ENGINEER & SUPPLY CO.,LTD. Renew coils Engineering & Supply Co., Ltd.	
Checked By	08-Aug-2012	Mr.Chatchawan S.		
Approved By	08-Aug-2012	Mr.Amnauy W.		
Certified By	08-Aug-2012	Mr.Amnauy W.		
Rev.	Date	Checked	Approved	Description
0	08-Aug-2012	Mr.Chatchawan S.	Mr.Amnauy W.	Issue for Approved

Renew Coils Engineering & Supply Co., Ltd.

Doc. No. : RNC-WPS-019-2012

Addendum

WELDING PROCEDURE SPECIFICATION (WPS)
RNC-WPS-010

RNC
RENEWCOILS ENGINEER & SUPPLY CO.,LTD.

WELDING PROCEDURE SPECIFICATION (WPS)

Company Name: Renew Coil Engineering & Supply Co., Ltd.

WPS No.: RNC-WPS-010 Date: 9-Jun-12 Revision No: 0

Supporting PQR No.(s): RNC-PQR-010 Date: 9-Jun-12 Type(s): Manual

Welding Process(es): Gas Tungsten Arc Welding (GTAW) (Automatic, **Manual**, Machine, or Semi-Auto.)

JOINT (QW-402)

Joint Design: **Single V**

Backing (Yes) ____ (No) X

Backing Material (Type): N/A

☐ Metal ☐ Nonfusing Metal
☐ Nonmetallic ☐ Other

Notes: ____

See Details Attachment sheet Joint Sketch

BASE METALS (QW-403)

P-No: 1 Group No.: 1 or 2 to P-No: 1 Group No: 1 or 2

Material Specification Type and Grade: A106 Gr.B, A234 WPB Equivalents TO A106 Gr.B, A234 WPB Equivalents

Chem. Analysis and Mech. Prop: -

Thickness Range:

Base Metal: Groove: 1.6 mm - 12.04 mm Fillet: All

Pipe Dia. Range: Groove: All Fillet: All

Other: ____

FILLER METALS (QW-404)	GTAW	
Spec. No. (SFA)	5.18	
AWS No. (Class)	ER 70S-G	
F-No	6	
A-No	1	
Size of Filler Metals	Ø 2.0 & 2.4 mm	
Deposited Weld Metals	6.02 mm	
Thickness Range:		
Groove	12.04 mm Max	
Fillet	All	
Electrode-Flux (Class)	-	
Flux Trade Name	-	
Consumable Insert	-	
Other	Kobelco TGS-50 or Equivalents	

POSITION (QW-405)

Position(s) of Groove: ALL

Welding Progression: Up X Down ____

Position(s) of Fillet: ALL

POSTWELD HEAT TREATMENT (QW-407)

Temperature Range: 593°-649°C

Time Range: Holding Time 1 Hour

Notes: Cooling Rate = 250° Max.

PREHEAT (QW-406)

Preheat Temp. Min: 10°C

Interpass Temp. Max: 250°C

Preheat Maintenance: N/A

Notes: ____

GAS (QW-408)

Percent Composition

	Gas(es)	(Mixture)	Flow Rate
Shielding	Argon	99.99%	10-20 Ltr/Min
Trailing	-	-	-
Backing	-	-	-

Sheet 1 of 2

RNC
REMEWCOILS ENGINEER & SUPPLY CO.,LTD

WELDING PROCEDURE SPECIFICATION (WPS)

WPS No. RNC-WPS-010 Rev. 0

ELECTRICAL CHARACTERISTICS (QW-409)

Current [ ] AC [X] DC Polarity EN

Amps (Range) 80-140 Volts (Range) 10-15

Tungsten Electrode Size and Type Ø 2.4 mm , Thoriated (EWTh-2)

(Pure Tungsten, 2% Thoriated, etc.)

Mode of Metal Transfer for GMAW N/A

(Spray arc, short circuiting arc, etc.)

Electrode Wire feed speed range N/A

TECHNIQUE (QW-410)

String or Weave Bead BOTH APPLICATIONS

Orifice or Gas Cup Size 8.0 mm - 19.0 mm

Initial and Interpass Cleaning (Brushing, Grinding, etc.) BRUSHING OR GRINDING

Method of Back Gouging N/A

Oscillation N/A

Contact Tube to Work Distance N/A

Multiple or Single Pass (per side) MULTIPASS

Multiple or Single Electrodes SINGLE ELECTRODE

Travel Speed (Range) 3.0 - 12.5 Cm/Minute

Peening N/A

Other N/A

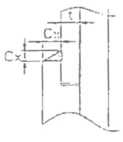
		Filler Metal		Current				
Weld Layer(s)	Process	Class	Dia. (mm)	Type Polar.	Amp. Range	Volt Range	Travel Speed Range (Cm/min)	Other
1st	GTAW	ER 70S-G	2.4	DC EN	80-100	10-12	3.0 -5.0	
2nd	GTAW	ER 70S-G	2.4	DC EN	100-120	12-13	3.0 -5.0	
Other(s)	GTAW	ER 70S-G	2.4	DC EN	120-140	13-15	7.0 -12.5	

We,the undersigned, certify that the statements in this record are correct and that the test welds were prepared ,welded, and tested in accordance with the requirement of **ASME-IX, 2007 latest Revision.**

	Name	Signed	Date
Prepared by :	Mr. Chatchawan.S		9/06/12
Reviewed by :	Mr. Chatchawan. S.		9/06/12
Approved by:	AMNUAY W.		20-08-55

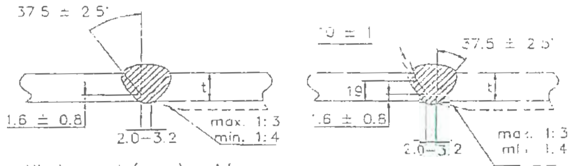
Sheet 2 of 2

ATTACHED FIGURE JOINT SKETCH



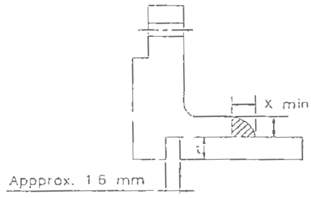
Cx (min.)=1.25t but not less than 3.2 mm

JOINT No. A1



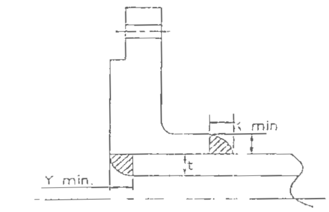
Misalignment (max.) – 1.5 mm

JOINT No B1 & B2



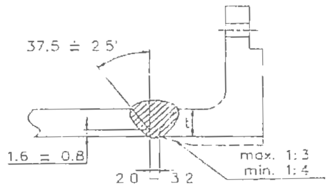
X min = The lesser of 1.4t or thickness of the hub.

JOINT No. A2



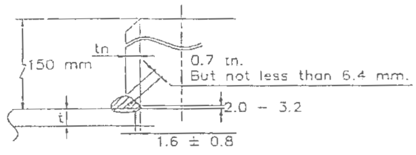
X min. = The lesser of 1.4t or thickness of the hub.
Y min. = The lesser of t or 6.4 mm.

JOINT No A3



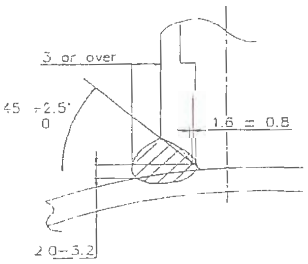
Misalignment (max.) = 1.5 mm

JOINT No B3

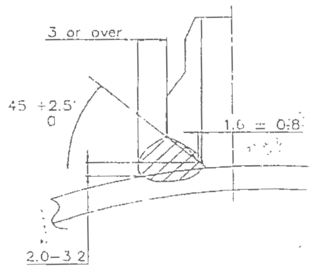


Misalignment (max.) = The lesser of 3.2 mm or 0.5tn.

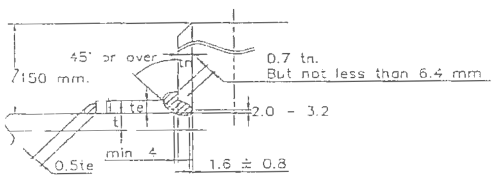
JOINT No. B4



JOINT No. C1

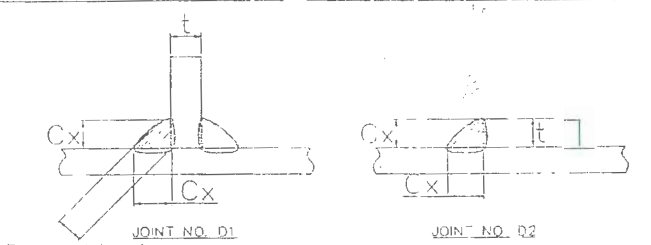


JOINT No. C2



Misalignment (max.) = The lesser of 3.2 mm. or 0.5tn.

JOINT No. B5



JOINT NO. D1 JOINT NO. D2

Theoretical throat

Note : Cx = Size of weld as specified
Theoretical throat = 0.707 x Cx

Procedure Qualification Record (PQR)

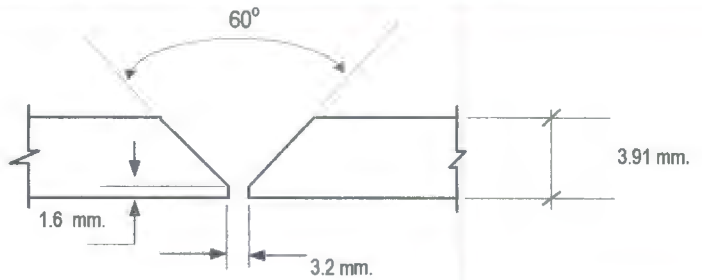
ASME-QW-200.2, Section IX, ASME Boiler Pressure Vessel Code

Record Actual Variables Used to Weld Test Coupon

RECORD OF WELDING (QW-483) PQR Number : PQR-RNC-010

Company Name :	RENEWCOILS ENGINEERING & SUPPLY CO.,LTD.		
PQR Record Number :	RNC-PQR-010	Date :	6 August 2012
WPS Number :	RNC-WPS-010		
Welding Process (es) :	Gas Tungsten Arc Welding (GTAW)		
Type (Manual, Semi-auto, or Automatic):	Manual		

JOINT DESIGN (QW-402)



(For combination qualifications, the deposited weld metal thickness shall be recorded for each filler metal or process used)

WELDING VARIABLES

BASE METALS (QW-403)

Material Spec.	A106	To	A106
Type or Grade	Gr.B	To	Gr.B
P. No.	1	To	1
Thickness of Test Coupon	6.02 mm.		
Diameter of Test Coupon	Ø 4"		
Others			

FILLER METALS (QW-404)

SFA Specification	A 5.18
AWS Classification	ER70S-G
Filler Metal F-No.	6
Weld Metal Analysis A-No.	1
Size of Filler Metal	Ø 2.4
Others	Kobelco TGS-50
Weld Metal Thickness	-

POSITION (QW-405)

Position of Groove	6G
Progression of Welding (Uphill or Down)	Up Hill
Others	N/A

PREHEAT (QW-406)

Preheat Temperature	Ambient Temperature
Interpass Temperature (max.)	234 °C
Others	N/A

POSTWELD HEAT TREATMENT (QW-407)

Temperature	610 °C
Hold Time	1 HR.
Others	Cooling Rate 180°C/Hr

GAS (QW-408)

	Percent Composition		
	Gas (es)	Mixture	Flow Rate
Shielding	Argon	99.95%	13 L/Min.
Trailing	-	-	-
Backing	-	-	-

ELECTRICAL CHARACTERISTICS (QW-409)

Current	DC		
Polarity	EN		
Amperage	80 - 120	Voltage	11 - 13
Tungsten Electrode Size	ø 2.4 mm. 2% thoriated.EWTh-2		
Others	N/A		

TECHNIQUE (QW-410)

Travel Speed	9.70 - 13.12 cm/min.
String or Weave Bead	Both
Oscillation	N/A
Multipass or single Pass	Multiple Pass
Multipass or Single Electrodes	Single Electrode
Others	N/A

PAE Technical Service Company Limited Public Registration No. (5) 1571/2542

69 Srinakarin Rd., suanluang, Bangkok 10250, Thailand. Tel (662) 322-022 Fax : (622) 721-2577

Procedure Qualification Record

ASME-QW-200.2, Section IX (WPQR)

RECORD OF TEST RESULTS (QW-483) PQR Number-: RNC-PQR-010

Tensile Test Report No.TS-12-08-078

Specimen	Width (mm)	Thickness (mm)	Area (mm^2)	Ultimate Total Load (KN)	Ultimate Strength (N/mm^2)	Type of Failure & Loc.
3177/12 TR1	19.43	5.78	112.305	58.173	517.991	OUT OF WELD
3177/12 TR2	19.23	5.98	114.995	58.565	509.283	OUT OF WELD

Guided Bend Tests (QW-160) Report No.BD-12-08-017

Type and Figure Number	Type of Bend	Indication	Results
3177/12 BF1	Transverse face bend	No Open Discontinuity	Accepted
3177/12 BF2	Transverse face bend	No Open Discontinuity	Accepted
3177/12 BR1	Transverse root bend	No Open Discontinuity	Accepted
3177/12 BR2	Transverse root bend	No Open Discontinuity	Accepted

Toughness Tests (QW-170)

Specimen Number	Notch Location	Notch Type	Test Temp.	Impact Values	Lateral Expansion		Drop Weight	
					% Shear	Mils	Break	No Break
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Fillet Weld Test (QW-180)

Result - acceptable : Yes N/A No N/A Penetration Into Base Metal : Yes N/A No N/A

Macroetch Test Results N/A

Other Tests

Type of Test Radiographic examination results Accepted Report No.PQR -013

Hardness Testing Report No. HN-12-08-030

Macrostructure Report No. MA-12-08-021

Impact Testing Report No.IM-12-08-083

Welders Name Mr.Choochat M. Welder Number -

Tests Conducted By Mr.Keerati Sa-ngoen Laboratory Test LPN PLATE MILL

We certify that the statements in this record are correct and that the test coupons were prepared, welded, and tested in accordance with the requirements of section IX of the ASME Boiler and Pressure Vessel Code.

Manufacture or Contractor :

Date : 19/08/12

Inspected by : Mr.Wichit K.

Certified by : Mr.Chaloemkiet R.

Date : 17-Aug-2012

PAE TECHNICAL SERVICE

PAE Technical Service Company Limited Public Registration No. (5) 1571/2542

69 Srinakarin Rd., suanluang, Bangkok 10250, Thailand. Tel (662) 322-022 Fax : (622) 721-2577

PAE TECHNICAL SERVICE CO.,LTD

NDT, TESTING, INSPECTION & ENGINERING CONSULTANT

69 SOI.OONNUT 64 (SUKSAMARN) SRINAKARIN RD, SUANLUANG,BANGKOK 10250

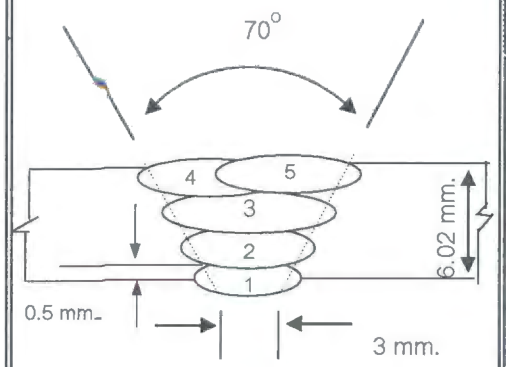
Tel : 02) 322-0222, Fax : 02)721-2577, RAYONG (038) 607-402-3

WELDER PROCEDURE QUALIFICATION RECORD

Client : Renew Coils Engineering & Supply Co.,Ltd.	**Project Name :** PQR	**Location :** RNC Work Shop	**Report No.** PQR-011
	Customer Order No. : N/A	**WPS No. :** RNC-WPS-010	**Date :** 6-Aug-12

Welder No.	Welder Name	Welding Process	Test Position	Applicable Standard	Material		Electrode Specification	Results	
					Spec	Size		Visual	RT
-	MR.CHOOCHAT M.	GTAW	6G	ASME - IX	A106 Gr.B	ø 4" x 6.02 mm.	A 5.18	ACCEPTED	ACCEPTED

Joint Details

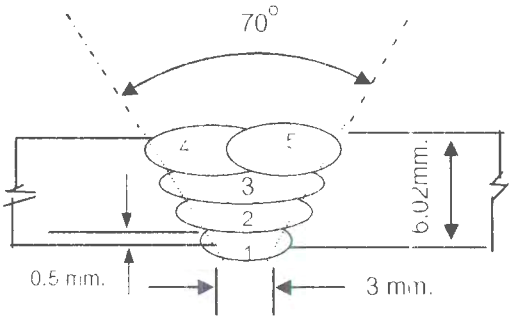


Layer (S)	Process	Filler Metal		Current		Volt	Travel Speed cm/min	Interpas Temp °C	Heat Input (KJ / cm)	Shielding		Backing	
		Class	Dia. (mm.)	Type Polarity	Amp					GAS	Flow Rate (l/min)	GAS	Flow Rate (l/min)
1	GTAW	ER 70S-G	2.4	DCEN	80	11	10.67	30	4.95	Argon	13	-	-
2	GTAW	ER 70S-G	2.4	DCEN	120	13	12.60	93	7.43	Argon	13	-	-
3	GTAW	ER 70S-G	2.4	DCEN	115	12	13.12	181	6.31	Argon	13	-	-
4	GTAW	ER 70S-G	2.4	DCEN	118	13	12.80	221	7.19	Argon	13	-	-
5	GTAW	ER 70S-G	2.4	DCEN	120	13	9.70	234	9.65	Argon	13	-	-

PAE Inspection By	Witness By	Approved By
SIGNATURE :	SIGNATURE :	SIGNATURE :
NAME : Mr.Wichit K	NAME : Mr.Chatchawan S	NAME : Mr.Chatchawan S
DATE : 7-Aug-2012	DATE : 7/08/12	DATE : 7/08/12

PAE TECHNICAL SERVICE	WELDING PROCEDURE	PQR No. : RNC-PQR-010
		WPS No. : RNC-WPS-010
	QUALIFICATION RECORDS	THIS PQR RECORDS ACTUAL TEST
		CONDITION AND RESULT

Welding Sequences for RNC-PQR-010 (Butt Joint)



Weld Bead No.	Process	Class Filter Metal	Wire Size (mm)	Arc Current (Amps)	Arc Voltage (Volts)	Pot. AC/DC+	String or Weave Bead	Wire Feed Speed	Rate of Travel (cm/min)	Heat Input (KJ/cm)	Gas Shielding Flow Rate (l/min)	Gas Backing Flow Rate (l/min)
1	GTAW	ER70S-G	2.4	80	11	DCEN	-	-	10.67	4.95	13	-
2	GTAW	ER70S-G	2.4	120	13	DCEN	-	-	12.60	7.43	13	-
3	GTAW	ER70S-G	2.4	115	12	DCEN	-	-	13.12	6.31	13	-
4	GTAW	ER70S-G	2.4	118	13	DCEN	-	-	12.80	7.19	13	-
5	GTAW	ER70S-G	2.4	120	13	DCEN	-	-	9.70	9.65	13	-

PAE Technical Service Company Limited
Public Registration No. (5) 1571/2542
69 Srinakarin Rd., suanluang, Bangkok 10250, Thailand. Tel (662) 322-022 Fax : (622) 721-2577

PAE TECHNICAL SERVICE

PAE TECHNICAL SERVICE CO.,LTD.

NDT, TESTING, INSPECTION & ENGINEERING CONSULTANT

TEL : 02) 322-0222 , FAX : 02) 721-2577 , RAYONG Tel : 038)607402-3, Fax : 038)607402, SRIRACHA : 038)330220, 330229

REPORT No.	PQR-013	**RADIOGRAPHIC EXAMINATION REPORT**	INSPECTION & TEST RECORDS
DWG.No.	PQR		
DATE :	6-Aug-12		

CLIENT :	Renew Coils Engineering & Supply Co.,Ltd.	PROJECT NAME :	PQR

PROCEDURE No. :	PAE-RT-001 (Rev.0)	EXPOSURE TIME		min. sec.	SOURCE TO FILM DISTANCE	4" Inch. / - mm.	
RADIATION SOURCE	X-ray (-) MAX - kVG. / γ - ray (Ir-192)	DENSITY RANGE		2.0 - 3.5	TYPE OF PENETRAMETER	ASTM-1A	
		SENSITIVITY		2%			
SOURCE RANGE	21.10 Ci.	FILM	TYPE	KODAK "MX"	FILM PROCESSING	AUTO	-
SOURCE SIZE	3 x 3 mm.		SIZE	3.5" x 8.5"		MANUAL	✓
TUBE VOLTAGE	- kVP.	INTENSIFYING SCREEN	FRONT	0.005 Inch.	MATERIAL	SA106 Gr.B	
TUBE CURRENT	- mA		BACK	0.005 Inch.	THICKNESS	6.02 mm.	

APPLICABLE STANDARD : ASME Section V Article 2 — ACCEPTANCE CRITERIA : ASME-IX

RADIOGRAPHIC TECHNIQUE : ☐ SWSI ☑ DWSI ☐ DWDI

NAME / WQT	Position No.	Pipe/Plate Size Dia	Pipe/Plate Size Thk(mm).	Welder No.	Film No.	Interpretation	Results	Remarks
GTAW / WPS No.RNC-PQR-010								
MR.CHOOCHAT M.	6G	4"	6.02	-	A-B	NAD	ACC	
					B-C	NAD	ACC	
					C-A	NAD	ACC	

Total	3.5"x8.5"	3	Films
	3.5"x17"	-	Films

ABBREVIATIONS :

ACC : Accepted	WH : Worm Hole	LP : Lack of Penetration	
REJ : Rejected	C : Crack	LF : Lack Of Fusion	CRP : Concave Root Penetration
CP : Cluster Porosity	P : Porosity	IP : Incomplete Penetration	ERP : Excess Root Penetration
SF : Surface Flaw	UC : Undercutting	TI : Tungsten Inclusion	NAD : No Apparent Defects
EUC : External Udercut	SL : Slag Inclusion	BT : Burn Through	RS : Reshoot

PAE INSPECTED BY		REVIEWED BY		APPROVED BY	
SIGNATURE :		SIGNATURE :		SIGNATURE :	
NAME :	Mr.Sarawut P.	NAME :	N. Chatchawan S	NAME :	
DATE :	7-Aug-12	DATE :	7/08/12	DATE :	

BOOK NO. 1149 NUMBER 24

HEAT

J.S.T. HEAT TREATMENT COMPANY LIMITED

POST WELD HEAT TREATMENT REPORT

CLIENT :	RENEW COILS ENGINEERING & SUPPLY CO.,LTD.	
CONTRACTOR :	J.S.T. HEAT TREATMENT CO.,LTD.	PAGE : 1
PROJECT :	PQR	
P.W.H.T.METHOD :	☑ LOCAL ○ FURNACE	
COMPLETION DATE :	08-08-2012	REPORT NO.PW-003
CHART SPEED :	☑ 25 MM/Hr, ○ 50 MM/Hr, ○ 100 MM/Hr.	
RECORDER NO :	EH-967Z014	MATERIAL : SA106 GR.B
LOCATION :	☑ SHOP ○ FIELD	UNIT :

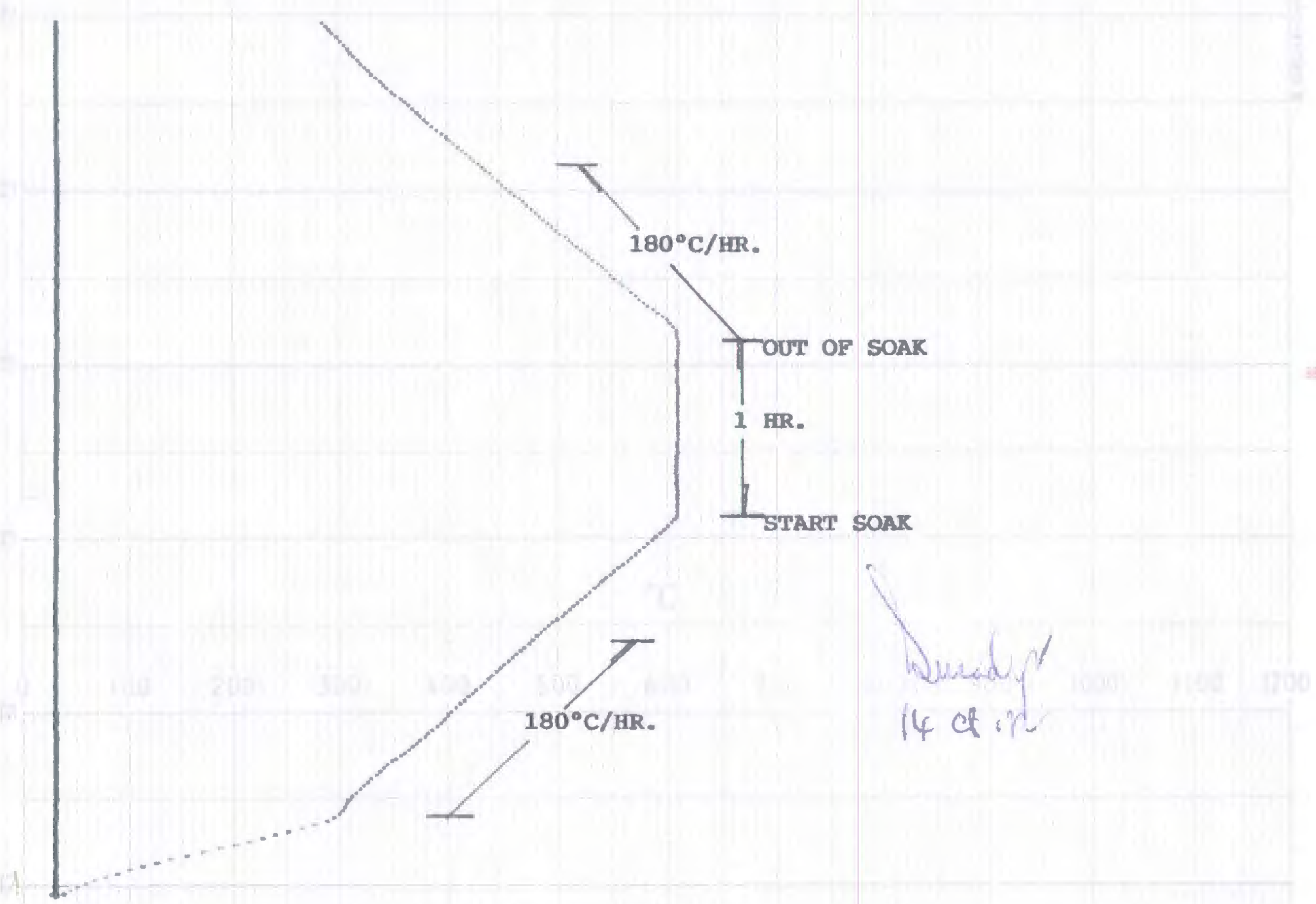
HEAT TREATMENT CONDITION/RESULTS

DESCRIPTION	LOADING TEMP(C°)	HEATING RATE (C°/Hr)	HOLDING TEMP (MIN-MAX)	HOLDING TIME (Hr)	COOLING RATE (C°/Hr)	UNLOADING TEMP (C°)
REQUEST	300	200	595-625	1 HR.	200	300
ACTUAL	300	180	610	1 HR.	180	300

DWG NO.,LINE NO.&CLASS,SHT.NO.SPOOL	WELD NO.	SIZE (")	THK. (MM)	WELD TYPE	T/C NO.
RNC-PQR-010		4"	6.02	BW	1,3

TOTAL	1 JOINTS	TOTAL	4 D.B.

	CHECKED BY J.S.T. HEAT TREATMENT CO.,LTD.	REVIEWED BY	APPROVED BY	APPROVED BY
SIGN :				
NAME :	SUPACHAI K.	Mr. Chatchawan S		
DATE :	08-08-2012	10/08/12	14 - 08 - 12	



DATE. 08-08-2012

RECORDER NO. EH - 967Z014

CHART NO. PW - 003

CHART SPEED.25mm./HR.

CUSTOMER. RENEW COILS ENGINEERING & SUPPLY CO.,LTD.

PROJECT. PQR

DWG NO. RNC-PQR-010

SIZE. 4" THK. 6.02mm.

MATERIAL. SA106 GR.B

START TIME. 08.00 AM.

TECHNICIAN. MR. MONTREE A.

TC	COLOUR
1.	RED
3.	SKY BLUE

QC

PAE TECHNICAL SERVICE

PAE TECHNICAL SERVICE CO.,LTD.

NDT, TESTING, INSPECTION & ENGINEERING CONSULTANT

TEL : 02) 322-0222 , FAX : 02) 721-2577 , RAYONG Tel : 038)607402-3, Fax : 038)607402, SRIRACHA : 038)330220, 330229

REPORT No.	PQR-017	RADIOGRAPHIC EXAMINATION REPORT	INSPECTION & TEST RECORDS
DWG.No.	PQR		
DATE :	8-Aug-12		

CLIENT :	Renew Coils Engineering & Supply Co.,Ltd.	PROJECT NAME :	PQR

PROCEDURE No. :	PAE-RT-001 (Rev.0)	EXPOSURE TIME		min. sec.	SOURCE TO FILM DISTANCE	4" Inch. / - mm.
RADIATION SOURCE	X-ray (-) MAX - kVG. / γ - ray (Ir-192)	DENSITY RANGE		2.0 - 3.5	TYPE OF PENETRAMETER	ASTM-1A
		SENSITIVITY		2%		
SOURCE RANGE	26.73 Ci.	FILM	TYPE	KODAK "MX"	FILM PROCESSING	AUTO -
SOURCE SIZE	3 x 3 mm.		SIZE	3.5" x 8.5"		MANUAL ✓
TUBE VOLTAGE	- kVP.	INTENSIFYING SCREEN	FRONT	0.005 Inch.	MATERIAL	SA106 Gr.B
TUBE CURRENT	- mA		BACK	0.005 Inch.	THICKNESS	6.02 mm.
APPLICABLE STANDARD :	ASME Section V Article 2				ACCEPTANCE CRITERIA :	ASME-IX

RADIOGRAPHIC TECHNIQUE : ☐ SWSI ☑ DWSI ☐ DWDI

NAME / WQT	Position No.	Pipe/Plate Size Dia	Thk(mm).	Welder No.	Film No.	Interpretation	Results	Remarks
GTAW / WPS No.RNC-PQR-010								
MR.CHOOCHAT M.	6G	4"	6.02	-	A-B	NAD	ACC	After PWHT
					B-C	NAD	ACC	
					C-A	NAD	ACC	

Total 3.5"x8.5" 3 Films

3.5"x17" - Films

ABBREVIATIONS :

ACC : Accepted	WH : Worm Hole	LP : Lack of Penetration	
REJ : Rejected	C : Crack	LF : Lack Of Fusion	CRP : Concave Root Penetration
CP : Cluster Porosity	P : Porosity	IP : Incomplete Penetration	ERP : Excess Root Penetration
SF : Surface Flaw	UC : Undercutting	TI : Tungsten Inclusion	NAD : No Apparent Defects
EUC : External Udercut	SL : Slag Inclusion	BT : Burn Through	RS : Reshoot

	PAE INSPECTED BY	REVIEWED BY	APPROVED BY
SIGNATURE :			
NAME :	Mr.Sarawut P.	Mr. Chatchawan S.	
DATE :	9-Aug-12	9/08/12	

Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415 , E-mail : qa@lpnpm.co.th

23850

Tension Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. TS-12-08-078

Revision No. 0

Test Product	PQR No. RNC-PQR-010	Material Characterization	Pipe
Project Name	N/A	Test Method	ASTM A370-10 (Prepared by ASME IX : 2010)
Project No.	N/A	Standard of Test	ASME Sec. IX : 2010
Material Specification	A106 Gr.B	Type of Test	Reduced Section Tension
Dimension	Ø 4" Sch 40 (6.02 mm Thk.)	Equipment's Capacity	200 TONS
Date Received	10 August 2012	Equipment / Serial No.	SHIMADZU UH200A / 32098926
Date Tested	14 August 2012	Ambient Temperature	23 ± 5 °C

Item No.	Test No.	Specimen Dimension					Yield		Ultimate		Elongation (%)	Remark
		Diameter (mm)	Thickness (mm)	Width (mm)	Area (mm^2)	Gauge Length (mm)	Load (kN)	Strength (N/mm^2)	Load (kN)	Strength (N/mm^2)		
1	3177/12 TR1	N/A	5.78	19.43	112.305	N/A	N/A	N/A	58.173	517.991	N/A	OUT OF WELD
2	3177/12 TR2	N/A	5.98	19.23	114.995	N/A	N/A	N/A	58.565	509.283	N/A	OUT OF WELD

The unit of N/mm^2 is equivalent to MPa

Additional details : WPS No. RNC-WPS-010

Welding Process : GTAW

Filler Metal/Electrode Spec't : ER-70S-G/TGS-50

RENEW COILS ENGINEERING & SUPPLY CO., LTD. RNC

Reported by : ....

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date ....16.. / ..08.. / ..12..

Approved by : ....

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date ..16.. / ..08.. / ..12..

LPN PLATE MILL

Notes 1. The above results are valid exclusively for tested samples as mentioned in this report.

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Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnpm.co.th

23855

TESTING No. 0158

Bend Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. BD-12-08-017

Revision No. 0

Test Product	PQR No. RNC-PQR-010	Material Characterization	Pipe
Project Name	N/A	Test Method	ASME Sec. IX : 2010
Project No.	N/A	Standard of Test	ASME Sec. IX : 2010
Material Specification	A106 Gr.B	Type of Jig	Guided-Bend Roller Jig
Dimension	Ø 4" Sch 40 (6.02 mm Thk.)	Diameter of Mandrel	24 mm
Date Received	10 August 2012	Bend Angle	180 Degree.
Date Tested	14 August 2012	Equipment / Serial No.	SHIMADZU UH200A/32098926

Item No.	Test No.	Type of Bend	Size of Specimen (ThickxWidthxLength)	Indication	Location of Indication	Remark
1	3177/12 BF1	Transverse face bend	5.81x19.20x163.10 mm	No Open Discontinuity	N/A	-
2	3177/12 BF2	Transverse face bend	5.95x19.29x164.03 mm	No Open Discontinuity	N/A	-
3	3177/12 BR1	Transverse root bend	5.97x19.01x162.25 mm	No Open Discontinuity	N/A	-
4	3177/12 BR2	Transverse root bend	5.99x19.14x162.06 mm	No Open Discontinuity	N/A	-

Additional details : WPS No. RNC-WPS-010

Welding Process : GTAW

Filler Metal/Electrode Spec't : ER-70S-G/TGS-50

RNC RENEW COILS ENGINEERING & SUPPLY CO., LTD. QC/QA

Reported by : ....

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date ....16../....08../..12....

Approved by : ....

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date ....16../....08../..12....

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Testing Laboratory LPN Plate Mill Public Company Limited

12744

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Tel. 0 2815 6400-9 ext. 1179, 1210, Direct Line Fax. 0 2815 6415 , E-mail : qa@lpnpm.co.th

Hardness Testing Report

Customer : RENEW COILS ENGINEERING & SUPPLY CO.,LTD.

Report No. HN-12-08-030

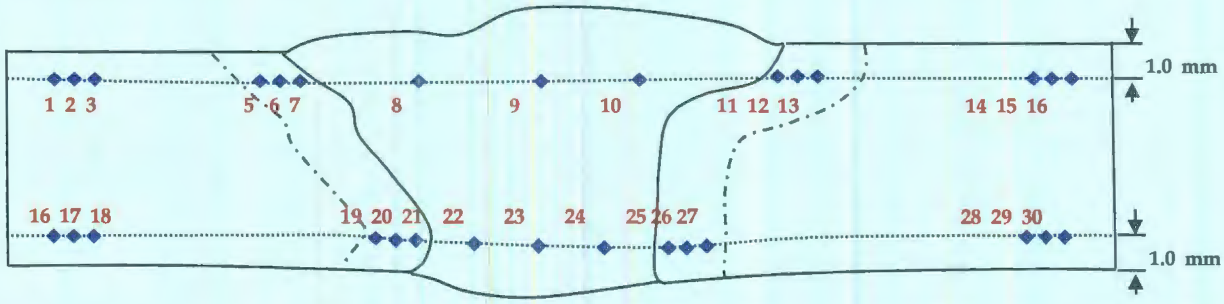
Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG,RAYONG 21000 Thailand.

Revision No. 0

Test Product	PQR No. RNC-PQR-010	Material Characterization	Pipe
Test No.	3177/12 HN	Test Method	ASTM E92-03
Project Name	N/A	Standard of Test	ASTM Sec. IX : 2010
Project No.	N/A	Type of test	Vicker Hardness Test
Material Specification	A106 Gr.B	Equipment	Mitutoyo, HV-115
Dimension	Ø4" Sch 40, (6.02 mm Thk.)	Indentor Size	Diamond, pyramid 136 deg.
Welding Process/Position	GTAW/All	Test Load	HV 10 Kgf.
Type of Joint	Butt joint	Total Force Dwell Time	15 s
Date Received	10 August 2012	Cal. Block	No. 62137 No. A62072
Date Tested	14 August 2012	Ambient Temperature	25.0 °C

Test Location

Total = 30 points



Point No.	Location	Hardness value Value (Scale HV)	Point No.	Location	Hardness value Value (Scale HV)	Point No.	Location	Hardness value Value (Scale)
1	BASE	149.5	16	BASE	150.1	******************************		
2	BASE	151.9	17	BASE	151.0			
3	BASE	151.2	18	BASE	151.4			
4	HAZ	151.6	19	HAZ	155.4			
5	HAZ	152.9	20	HAZ	158.1			
6	HAZ	154.9	21	HAZ	159.6			
7	WELD	159.8	22	WELD	156.5			
8	WELD	170.5	23	WELD	146.7			
9	WELD	174.9	24	WELD	145.7			
10	HAZ	161.2	25	HAZ	151.2			
11	HAZ	156.1	26	HAZ	150.2			
12	HAZ	150.6	27	HAZ	150.9			
13	BASE	159.5	28	BASE	153.6			
14	BASE	158.4	29	BASE	154.6			
15	BASE	158.1	30	BASE	154.7			

Additional details : Filler Metal : ER-70S-G TGS-50(GTAW)

RNC RENEW COILS ENGINEERING & SUPPLY CO., LTD. QC/QA RNC

Reported by : ....

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurancer

Date : 15/08/12

Approved by : ....

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date : 15/08/12

LPN PLATE MILL

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Tel. 0 2815 6400-9 ext. 1179, 1210, Direct Line Fax. 0 2815 6415 , E-mail : qa@lpnpm.co.th

1622

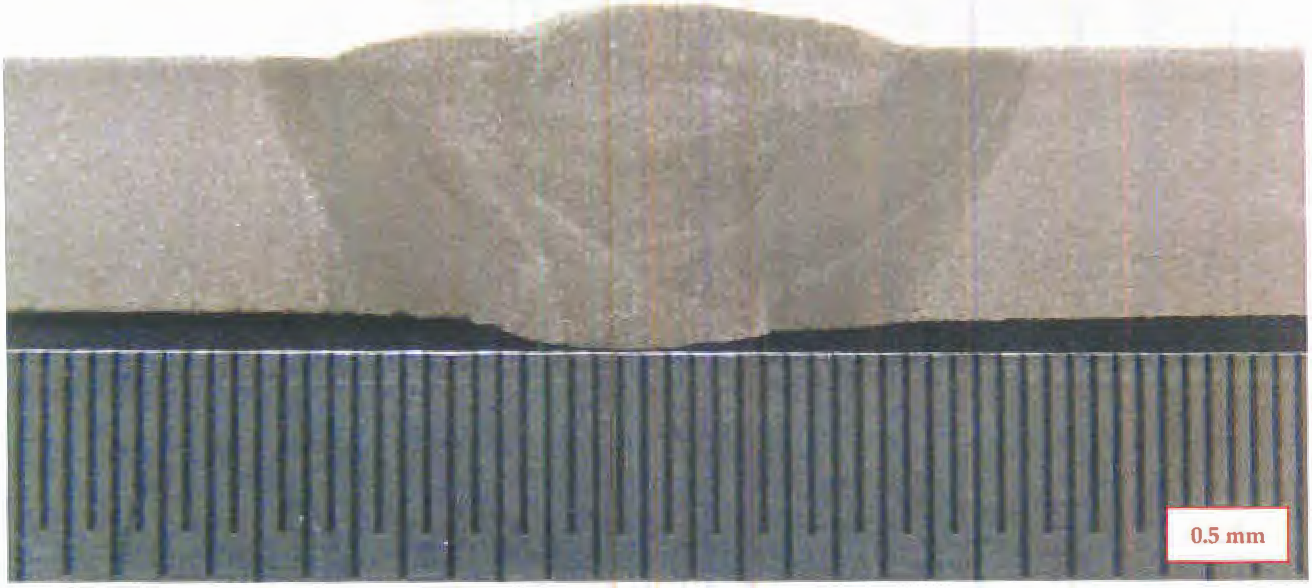
Macrostructure Analysis Report

Report No. MA-12-08-021
Revision No. 0

Customer : RENEW COILS ENGINEERING & SUPPLY CO.,LTD.
Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG,RAYONG 21000 Thailand.

Test Product	PQR No. RNC-PQR-010
Test No.	3177/12 MA
Project Name	N/A
Project No.	N/A
Material Specification	A106 Gr.B
Dimension	Ø4" Sch 40, (6.02 mm Thk.)
Welding Process/Position	GTAW/All
Type of Joint	Butt joint
Date Received	10 August 2012

Material Characterization	Pipe
Test Method	ASME E340-00
Standard of Test	ASME Sec. IX : 2010
Etchant Reagent	75ml H_2O+25ml HNO_3(65%)
Temperature	28 °C
Time of Etching	10 minutes
Surface Preparation	Polishing with sand paper 240, 400, 600, 800, 1000
Date Tested	11 August 2012



Additional details : The macrostructure shows complete fusion of the weldment. Weld metal and heat affected zone are free of cracks.
Filler Metal : ER-70S-G TGS-50(GTAW)

RENEW COILS ENGINEERING & SUPPLY CO.,LTD. RNC QC/QA

Reported by : ....
(Mr.Keerati Sa-ngoen)
(Acting) General Manager - Quality Assurancer
Date : 15/08/12

Approved by : ....
(Mr.Pavaret Preedawiphat)
Deputy Managing Director
Date : 15/08/12

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Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnpm.co.th

TESTING No 0158 23860

Impact Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. IM-12-08-083

Revision No. 0

Test Product	PQR No. RNC-PQR-010	Material Characterization	Pipe
Project Name	N/A	Test Method	ASTM A370-10 and ASTM E23M-07a$^{\varepsilon1}$
Project No.	N/A	Standard of Test	ASME Sec. IX : 2010
Material Specification	A106 Gr.B	Type of Test	Charpy V-Notch Impact
Dimension	Ø 4" Sch 40 (6.02 mm Thk.)	Equipment / Serial No.	Tinius Olsen / 221811
Date Received	10 August 2012	Equipment 's Capacity	542 Joules
Date Tested	15 August 2012	Ambient Temperature	23 ± 5 °C

Item No.	Test No.	Test Temperature (°C)	Size of Specimen			Results		Location of Test	Remark
			Thickness (mm)	Width (mm)	Length (mm)	Energy Absorbed (Joules)	Average (Joules)		
1	3177/12-1	-20°C	5.011	10.004	54.75	78.76	82.89	Base Metal	-
2	3177/12-2	-20°C	5.021	10.009	54.75	85.59			
3	3177/12-3	-20°C	5.016	10.015	54.77	84.33			
4	3177/12-1	-20°C	5.018	10.009	54.90	129.32	109.53	Weld Metal	-
5	3177/12-2	-20°C	5.021	10.007	54.93	98.18			
6	3177/12-3	-20°C	5.015	10.009	54.96	101.10			
7	3177/12-1	-20°C	5.014	10.008	54.99	93.65	90.20	HAZ	-
8	3177/12-2	-20°C	5.009	10.008	54.99	92.26			
9	3177/12-3	-20°C	5.012	10.008	54.98	84.68			

Additional details : WPS No. RNC-WPS-010

Welding Process : GTAW

Filler Metal/Electrode Spec't : ER-70S-G/TGS-50

RNC RENEW COILS ENGINEERING & SUPPLY CO., LTD. QC/QA RNC

Reported by : ....

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date ....16..../....08..../....12....

Approved by : ....

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date ....16..../....08..../....12....

LPN PLATE MILL

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KOBELCO

INSPECTION CERTIFICATE

Date of Issue	17/Feb/2011
Certificate No.	KF11-002
Date of Inspection	15/Feb/2011

PURCHASER

TRADE DESIGNATION	SIZE (mm)	MFG. No.	APPLICABLE SPECIFICATION & CLASSIFICATION
TG-S50	2.4	F0B2704	AWS A5.18 ER70S-G

Chemical Composition (%)

Elements	C	Si	Mn	P	S	Cu	Ni	Cr	Mo	V
WIRE	0.10	0.75	1.39	0.011	0.017	0.13	0.01	0.01	< 0.01	0.002

Tensile Test of Weld Metal			Impact Test of Weld Metal			
Yield Strength at 0.2% offset	Tensile Strength	Elongation	Test Temp.	Absorbed Energy		Tested size for Mechanical test: 2.4 mm Tested MFG. No. of
453 MPa	545 MPa	38 %	-30 °C	Avg.	257 J	Mechanical test: F9B4140

Welding Conditions				REMARKS
Type of Current	DCEN	Shielding Gas	Ar	
Welding Current	226 A	----------	----------	
Welding Voltage	14 V	----------	----------	

We hereby certify that the test results of the above welding material are as described herein and satisfy the requirements of applicable specification.

KOBE MIG WIRE (THAILAND) CO., LTD.

CHIEF INSPECTOR

ArcelorMittal South Africa Limited
Tubular Products
Genl. Hertzog rd.
P O Box 48 Vereeniging 1930
South Africa

MATERIAL TEST CERTIFICATE
SEAMLESS TUBE

Telephone +27 (0)16 450 4220
Fax +27 (0)16 423 4906

Licensed under API Spec 5L and 5CT
5L - 0319
5CT - 0419

ISO 9001: 2008
DE-395997 QM
Test Certificate EN 10204:2004 TYPE 3.1

ArcelorMittal

Customer:	UR MENA LIMITED TE-4137 ITEM: 14	Customer Order/Contract No:	STH2 138413
Order No:	4000013957	Material No:	1000000402
Certificate Reference No:	040060580242	Cast/Heat No:	98B5502
Product:	FULLY KILLED HOT FINISHED CARBON STEEL SEAMLESS TUBES		Page 1 of 1
Specification:	ISO3183:2007/API 5L:2007 L245/L290/B/X42 PSL1 ASTM A106.08B/A53B.07/ A530.04 ASME SA106.08B/SA53B.07/SA530.04		
Product Marking:	ARCELORMITTAL SA ISO3183/Spec 5L-0319 MONOGRAM 02-12 ASTM/ASME A/SA106B A/SA53B 114.300 (4.500) 6.020 (0.237) 6.000 (19.700) L245/B L290/X42 PSL1 SMLS TESTED 20.7 MPa (3000 psi) CAST NO:98B5502 PROD/O NO: T8341395710		

General Information

Quantity	Mass	Dimensions			Total Length
		Tube OD	Thickness	Length	
71	6,847.950(kg)	114.300(mm)	6.020(mm)	6.000(m)	426.000(m)
	15,096.991(lb)	4.500(")	0.237(")	19.700 (ft)	1,398.700(ft)

Steel making process	Final Rolling Operation
Electric Arc	Final hot rolling operation finished above 860°C and cooled in still air. ✓

Chemical Composition

R22-(V+Nb+Ti) R24-(Nb+V)

Element(%)	C	Si	Mn	S	P	Cr	Ni	Mo	Cu	V	Al	Ti	Sn	Ca	N	B	Nb	CE	R22	R24
Minimum	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Maximum	0.200	-	1.30	0.030	0.030	0.50	0.50	0.150	0.500	-	-	-	-	-	-	-	-	0.41	0.15	0.060
Heat	0.189	0.30	0.82	0.006	0.008	0.09	0.05	0.008	0.090	0.002	0.043	0.001	0.007	0.0031	0.0095	0.0002	0.0010	0.36	0.00	0.003
Product	0.189	0.300	0.820	0.006	0.008	0.090	0.050	0.008	0.090	0.002	0.043	0.001	0.007	0.003	0.009	0.000	0.001	0.355	0.004	0.003
Product (ADD)	0.189	0.300	0.820	0.006	0.008	0.090	0.050	0.008	0.090	0.002	0.043	0.001	0.007	0.003	0.009	0.000	0.001	0.355	0.004	0.003

pttGC GLOBAL CHEMICAL
Chayut S.
Inspection QA/QC Division
Integrity and Reliability
Chayut

Mechanical Properties

Specification Limits	UTS (Rm) MPa	UTS (Rm) psi	Yield (0.5%) MPa	Yield (0.5%) psi	% EL 50mm
Minimum	415	60000	290	42000	30.0
Maximum	-	-	-	-	-
(1) Actual	491	71213	331	48007	38.1
(2) Actual	495	71793	337	48877	39.1
(3) Actual					
(4) Actual					

	UTS (Rm) MPa	UTS (Rm) psi	Yield (0.5%) MPa	Yield (0.5%) psi	% EL 50mm
(5) Actual					
(6) Actual					
(7) Actual					
Orientation & type of tensile test			Longitudinal, Strip		
Width of tensile piece (mm)			25.4mm		
Orientation of impact test piece					

OTHER TESTS	
Flattening	Passed
Hydrostatic	20700 kPa (3000 psi) - 5 sec
NDI: EMI	PASS - ASTM E570 - 12.5% NOTCH
NDI: UT	UT-not required
HV 10 Kg	151 154 157 ✓

A50155710
3

UR MENA LIMITED Q.A.

Remarks:

Material in accordance with NACE MR0175:2003/ISO15156-2:2003, MR0103:2010. Dimensions to ANSI B36.10M-2004
The material conform to the hot yield strength requirements as per ASME, Sect II, Pt D, Table Y-1, 2010
All the material conform to the visual and dimensional requirements and is made to a suitable fine grain practice

Quality Assurance Manager: R. Bester *Date of Issue: 2012.03.08* *Certified by:*